

GURTEEN, Ireland

WMO: 039660

Lat: 53.017N

Lon: 8.000W

Elev: 76

StdP: 100.42

Time Zone: 1.00 (IRL)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.7	-2.8	-5.7	2.3	-3.2	-4.3	2.7	-1.6	13.6	8.2	12.5	8.2	1.4	40	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	23.1	17.2	21.2	16.5	19.7	15.7	18.1	21.7	17.2	20.2	16.4	19.0	3.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.0	19.5	16.1	11.5	18.6	15.2	10.9	17.7	51.3	21.9	48.7	20.1	46.3	19.1	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	10.0	8.8	-6.2	25.6	2.7	2.3	-8.1	27.2	-9.7	28.5	-11.2	29.8	-13.2	31.5		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-6.4	19.6	2.7	1.2	-8.4	20.5	-10.0	21.1	-11.5	21.8	-13.5	22.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	4.9	5.2	6.0	8.3	10.9	13.6	15.4	12.9	10.2	6.7	5.4
		DBStd	4.68	3.20	3.05	2.85	2.68	2.72	2.39	2.19	1.84	2.27	2.75	3.16
		HDD10.0	792	159	136	127	63	21	2	0	3	32	105	145
		HDD18.3	3218	417	368	382	301	231	143	94	119	163	253	348
		CDD10.0	621	1	2	3	12	48	110	168	140	90	37	7
		CDD18.3	6	0	0	0	0	0	1	4	1	0	0	0
		CDH23.3	59	0	0	0	0	2	23	34	1	0	0	0
		CDH26.7	8	0	0	0	0	0	6	2	0	0	0	0
(14)	Wind	WSAvg	4.2	4.7	4.6	4.4	4.3	4.2	3.7	3.6	4.0	4.1	4.1	4.3
(15)	Precipitation	PrecAvg	943	97	70	65	66	64	70	78	80	71	102	97
(16)		PrecMax	1129	163	182	113	115	141	132	142	172	154	160	225
(17)		PrecMin	736	30	16	19	16	7	18	43	9	25	28	49
(18)		PrecStd	103	36	45	26	29	32	33	24	43	43	40	42
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	13.0	13.1	15.9	18.6	23.2	27.0	26.8	22.2	21.6	18.0	14.9	13.7
(20)		0.4% MCWB	11.9	10.8	10.5	13.7	17.1	17.1	19.2	17.4	17.2	15.8	13.7	12.3
(21)		2% DB	11.5	11.8	13.4	16.3	20.2	22.5	23.7	20.3	19.4	16.4	13.8	12.7
(22)		2% MCWB	10.6	10.3	10.4	12.0	15.5	16.3	17.5	16.5	15.9	14.7	12.6	11.4
(23)		5% DB	10.7	11.0	11.9	14.4	17.8	20.4	21.7	19.1	18.0	15.4	12.8	11.7
(24)		5% MCWB	9.8	9.8	9.6	10.9	13.4	15.5	16.8	15.6	15.2	13.8	11.6	10.6
(25)		10% DB	9.8	10.1	10.8	12.9	15.9	18.7	19.8	18.1	16.8	14.5	11.7	10.9
(26)		10% MCWB	8.8	9.0	8.9	10.1	12.4	14.7	15.8	15.1	14.4	13.0	10.6	10.0
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	12.0	11.2	11.7	14.3	17.5	18.2	19.9	18.4	18.1	16.2	14.1	12.7
(28)		0.4% MCDB	12.7	12.2	13.7	18.1	22.1	23.9	25.2	20.9	20.5	17.3	14.7	13.6
(29)		2% WB	10.8	10.7	10.7	12.6	15.9	17.1	18.5	17.4	16.7	15.0	12.7	11.6
(30)		2% MCDB	11.4	11.6	12.6	15.4	19.9	21.8	22.5	19.5	18.5	16.0	13.5	12.5
(31)		5% WB	10.0	10.0	10.0	11.4	14.3	16.1	17.5	16.5	15.8	14.1	11.8	10.9
(32)		5% MCDB	10.6	11.1	11.8	13.9	17.0	19.5	20.5	18.5	17.4	15.3	12.7	11.7
(33)		10% WB	9.0	9.2	9.1	10.5	13.0	15.1	16.5	15.6	14.9	13.2	10.7	10.0
(34)		10% MCDB	9.8	10.1	10.8	12.7	15.1	18.1	19.1	17.7	16.5	14.3	11.7	11.0
(35)	Mean Daily Temperature Range	MDBR	5.3	6.0	7.6	8.2	8.2	8.3	7.7	7.1	7.1	6.5	5.8	5.4
(36)		5% DB MCDBR	5.7	5.6	8.5	10.0	11.2	12.1	10.8	8.0	8.2	6.1	5.5	5.7
(37)		5% WB MCWBR	5.5	4.9	6.2	6.7	6.6	7.0	5.8	4.9	5.6	4.9	5.1	5.4
(38)		5% WB MCDBR	5.8	5.5	8.0	9.1	9.9	10.4	9.2	7.1	6.8	5.8	5.5	5.6
(39)	Clear-Sky Solar Irradiance	taub	0.339	0.348	0.371	0.379	0.390	0.376	0.380	0.362	0.356	0.345	0.332	0.329
(40)		taud	2.237	2.292	2.259	2.271	2.277	2.348	2.344	2.407	2.419	2.402	2.294	2.204
(41)		Ebn at Noon	613	744	806	851	859	875	863	862	820	744	621	541
(42)		Edh at Noon	67	85	106	119	124	117	116	103	90	75	62	57
(44)	All-Sky Solar Radiation	RadAvg	0.71	1.40	2.50	3.88	4.83	5.07	4.58	3.88	2.86	1.68	0.88	0.54
(45)		RadStd	0.07	0.12	0.25	0.37	0.33	0.46	0.38	0.30	0.19	0.14	0.08	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page